

Assignment Week 3 – Bioinformatics File Formats

1. What type of data is stored in a FASTA file?
2. Why do we use BAM instead of SAM?
3. What do REF and ALT mean in a VCF file?

Answers:

1. A FASTA file contains DNA, RNA or protein sequences along with a header line which describes what sequences they are or which organism or gene they belong to.
2. BAM file which stands for Binary Alignment map is used in place of sequence alignment map (SAM) file because of size. SAM files are usually large in size hence occupy more disk or storage space. As a result, these SAM files are converted into binary format known as BAM files which are used as input for variant calling leading to their conversion into VCF file format. This conversion also occurs faster using the bam files.
3. The REF in the VCF file format refers to the reference allele (In the reference genome) whereas the ALT refers to the alternate allele (that is the allele represented in our sequences) matched against the reference genome or sequence.